Lecture 27

Invariant tensors and commutants

Objectives. *This lecture has two aims. The first is to identify intertwiners with invariant tensors. This gives commutants from multiplicity spaces, explains classical contractions and gives the projectors of a vector square. The second aim is to determine the representation ring, in which characters turn tensor products into multiplication.*

By the end of this lecture you will be able to:

- Translate between intertwiners, invariant tensors and invariant bilinear forms.
- Compute a commutant from multiplicity spaces.
- Distinguish O_N-invariants from $\mathfrak{so}_N$-invariants, including volume tensors.
- Prove that the representation ring is a polynomial ring in the fundamental characters.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra, and modules are finite-dimensional. The named classical examples also use $\mathfrak{gl}_n$, the zero algebra $\mathfrak{sl}_1$, the abelian algebra $\mathfrak{so}_2$ and matrix groups. Recall the dual and tensor actions from Lecture 14:

$$(x\xi)(u) = -\xi(xu), \qquad x(u \otimes w) = xu \otimes w + u \otimes xw.$$

The *invariants* of a module W are

$$W^{\mathfrak{g}} = \{w \in W : xw = 0 \text{ for all } x \in \mathfrak{g}\} \cong \mathrm{Hom}_{\mathfrak{g}}(\mathbb{C}, W).$$

27.1 Intertwiners as invariant tensors

Proposition 27.1 (Intertwiners and invariants). *Let U and V be modules. The linear isomorphism*

$$\Phi\colon U^* \otimes V \longrightarrow \mathrm{Hom}(U,V), \qquad \Phi(\xi \otimes v)(u) = \xi(u)\, v,$$

satisfies

$$\Phi(xt) = \rho_V(x)\Phi(t) - \Phi(t)\rho_U(x) \quad \text{for all } x \in \mathfrak{g},\ t \in U^* \otimes V.$$

Hence $\mathrm{Hom}_{\mathfrak{g}}(U,V) \cong (U^* \otimes V)^{\mathfrak{g}}$.

Proof. The map Φ is bijective because U and V are finite-dimensional. Both sides are linear in t, so it suffices to take $t = \xi \otimes v$. For $u \in U$, the tensor action, the dual action and the definition of Φ give

$$\begin{aligned} \Phi\big(x(\xi \otimes v)\big)(u) &= \Phi(x\xi \otimes v + \xi \otimes xv)(u) \\ &= -\xi(xu)\, v + \xi(u)\, xv \\ &= \big(\rho_V(x)\Phi(\xi \otimes v) - \Phi(\xi \otimes v)\rho_U(x)\big)(u). \end{aligned}$$

So $xt = 0$ for all x exactly when $\Phi(t)$ intertwines the actions. ∎

For $U = V$, the map Φ turns the evaluation $\xi \otimes v \mapsto \xi(v)$ into the trace:

$$\xi(v) = \operatorname{tr} \Phi(\xi \otimes v).$$

Let (v_j) be a basis of V with dual basis (v_j^*). The invariant tensor $\sum_j v_j^* \otimes v_j$ corresponds to id_V, and its evaluation is $\dim V$.

Corollary 27.2 (Tensor–Hom reciprocity)**.** *For modules U, V, W,*

$$\operatorname{Hom}_{\mathfrak{g}}(U \otimes W, V) \cong \operatorname{Hom}_{\mathfrak{g}}(U, W^* \otimes V).$$

Proof. The linear isomorphism $U^* \otimes W^* \to (U \otimes W)^*$, $(\xi \otimes \eta)(u \otimes w) = \xi(u)\eta(w)$, is equivariant:

$$(x\xi \otimes \eta + \xi \otimes x\eta)(u \otimes w) = -\xi(xu)\eta(w) - \xi(u)\eta(xw) = -(\xi \otimes \eta)\big(x(u \otimes w)\big).$$

Hence this isomorphism and Proposition 27.1, applied twice, give

$$\begin{aligned} \operatorname{Hom}_{\mathfrak{g}}(U \otimes W, V) &\cong \big((U \otimes W)^* \otimes V\big)^{\mathfrak{g}} \\ &\cong \big(U^* \otimes (W^* \otimes V)\big)^{\mathfrak{g}} \\ &\cong \operatorname{Hom}_{\mathfrak{g}}(U, W^* \otimes V). \end{aligned}$$ ∎

A bilinear form B on V is *invariant* if

$$B(xu, v) + B(u, xv) = 0 \quad \text{for all } x \in \mathfrak{g},\ u, v \in V.$$

Equivalently, the linear map $V \otimes V \to \mathbb{C}$, $u \otimes v \mapsto B(u, v)$, is an intertwiner to the trivial module $\mathbb{C}$.

Corollary 27.3 (Invariant bilinear forms)**.** *Let L be irreducible. The invariant bilinear forms on L form a space of dimension one if $L \cong L^*$, and zero otherwise. A nonzero invariant form is nondegenerate and is either symmetric or alternating.*

Proof. Reciprocity with $U = W = L$ and $V = \mathbb{C}$ gives

$$\operatorname{Hom}_{\mathfrak{g}}(L \otimes L, \mathbb{C}) \cong \operatorname{Hom}_{\mathfrak{g}}(L, L^*), \qquad B \longmapsto \big(u \mapsto B(u, \cdot)\big).$$

Schur's lemma gives the dimension of $\operatorname{Hom}_{\mathfrak{g}}(L, L^*)$. A nonzero form corresponds to an isomorphism $L \to L^*$, so it is nondegenerate.

The transpose $B^t(u, v) = B(v, u)$ is again invariant. If $L \cong L^*$, the involution $B \mapsto B^t$ acts on the one-dimensional space of invariant forms by $+1$ or -1. So a nonzero invariant form is symmetric or skew-symmetric, and over $\mathbb{C}$ a skew-symmetric form is alternating. ∎

27.2 Commutants from multiplicity spaces

By Lecture 14, a module V has multiplicity spaces $M_\lambda = \operatorname{Hom}_{\mathfrak{g}}(L(\lambda), V)$, of dimension m_λ, and the evaluation

$$\mathrm{ev}\colon \bigoplus_{\lambda \in P_+} M_\lambda \otimes L(\lambda) \longrightarrow V, \qquad \varphi \otimes u \longmapsto \varphi(u),$$

is an isomorphism. Here $\mathfrak{g}$ acts only on the factors $L(\lambda)$.

Theorem 27.4 (Isotypic commutant formula). *The map*

$$\Psi\colon\ \mathrm{End}_{\mathfrak{g}}(V) \longrightarrow \bigoplus_{\lambda} \mathrm{End}(M_\lambda), \qquad \Psi(T)_\lambda(\varphi) = T \circ \varphi,$$

is an algebra isomorphism. Hence

$$\dim \mathrm{End}_{\mathfrak{g}}(V) = \dim (V^* \otimes V)^{\mathfrak{g}} = \sum_{\lambda} m_\lambda^2.$$

Proof. If φ intertwines, so does $T \circ \varphi$. Thus $\Psi(T)_\lambda \in \mathrm{End}(M_\lambda)$. The map Ψ is linear and respects composition and identities, because $(ST) \circ \varphi = S \circ (T \circ \varphi)$.

Injectivity. If $T \circ \varphi = 0$ for all $\varphi \in M_\lambda$ and all λ, then T vanishes on every image $\varphi(L(\lambda))$. These images span V because ev is onto. Hence $T = 0$.

Surjectivity. Let $A_\lambda \in \mathrm{End}(M_\lambda)$. The operator $\bigoplus_\lambda A_\lambda \otimes \mathrm{id}$ commutes with $\mathfrak{g}$, since $\mathfrak{g}$ acts only on the second factors. Put

$$T = \mathrm{ev} \circ \Big(\bigoplus_{\lambda} A_\lambda \otimes \mathrm{id} \Big) \circ \mathrm{ev}^{-1} \in \mathrm{End}_{\mathfrak{g}}(V).$$

For $\varphi \in M_\lambda$ and $u \in L(\lambda)$,

$$T(\varphi(u)) = T\big(\mathrm{ev}(\varphi \otimes u)\big) = \mathrm{ev}(A_\lambda \varphi \otimes u) = (A_\lambda \varphi)(u).$$

So $T \circ \varphi = A_\lambda \varphi$, and $\Psi(T) = (A_\lambda)_\lambda$. The dimension formula follows, since $\dim \mathrm{End}(M_\lambda) = m_\lambda^2$. Proposition 27.1 gives the middle term. ∎

Hence the commutant $\mathrm{End}_{\mathfrak{g}}(V)$ is commutative exactly when every $m_\lambda \le 1$, that is, when V is *multiplicity-free*. The invariants of V form the multiplicity space of the trivial module: $V^{\mathfrak{g}} \cong M_0$.

For $V = \mathbb{C}^n$ and $\sigma \in S_d$, the *permutation operator* is

$$\sigma(v_1 \otimes \cdots \otimes v_d) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(d)}.$$

Corollary 27.5 (Commutant in type A). *Let $V = \mathbb{C}^n$ and $d \ge 1$. The commutants of $\mathfrak{gl}_n$ and of $\mathfrak{sl}_n$ on $V^{\otimes d}$ both equal the span of the permutation operators. Moreover,*

$$\big(V^{\otimes k} \otimes V^{*\otimes l}\big)^{\mathfrak{gl}_n} = 0 \quad \text{for } k \neq l.$$

Proof. For $n = 1$, every tensor space is one-dimensional. Permutations act as the identity, so their span is the commutant of either algebra. The mixed-tensor assertion follows since I acts by $k - l$. Assume $n \ge 2$. Since $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C} I$ and the identity matrix I acts on $V^{\otimes d}$ by d id, the two commutants agree. The finite-group course gives the Schur–Weyl decomposition

$$V^{\otimes d} \cong \bigoplus_{\lambda \vdash d,\ \ell(\lambda) \le n} \mathbb{S}_\lambda(V) \otimes V^\lambda.$$

Here V^λ is the Specht module of S_d, and S_d acts only on these factors.

By Lecture 18, the $\mathbb{S}_\lambda(V)$ are irreducible $\mathfrak{sl}_n$-modules, and two of them have the same highest weight only if the partitions differ by full columns of length n. Two different partitions of d cannot differ in this way, so the $\mathbb{S}_\lambda(V)$ are pairwise nonisomorphic. Their multiplicity spaces, indexed here by partitions, are $M_\lambda \cong V^\lambda$. The permutation σ acts on M_λ through its action on V^λ.

Theorem 27.4 gives

$$\operatorname{End}_{\mathfrak{gl}_n}(V^{\otimes d}) \cong \bigoplus_{\ell(\lambda)\le n} \operatorname{End}(V^\lambda).$$

By Wedderburn's theorem for $\mathbb{C}[S_d]$, the map $\mathbb{C}[S_d] \to \bigoplus_{\lambda\vdash d} \operatorname{End}(V^\lambda)$ is onto. So is its projection onto the summands with $\ell(\lambda) \le n$. Hence the permutation operators span $\operatorname{End}_{\mathfrak{gl}_n}(V^{\otimes d})$.

Finally, the identity matrix acts on $V^{\otimes k} \otimes V^{*\otimes l}$ by $(k-l)$ id. An invariant vector is therefore zero when $k \neq l$. ∎

For $k = l$, Proposition 27.1 turns σ into an invariant tensor that pairs the copies of V^* with the copies of V according to σ. For $k = 1$ this tensor is $\sum_i e_i^* \otimes e_i$, the tensor of id_V from Section 27.1.

27.3 Contractions, forms and volume tensors

Let B be a nondegenerate invariant bilinear form on V.

Lemma 27.6 (The invariant tensor of a form). *Let (v_j) be a basis of V, and choose v_j' with $B(v_j', v_k) = \delta_{jk}$. Then*

$$\Omega_B = \sum_j v_j \otimes v_j'$$

is invariant and independent of the basis.

Proof. The map $\phi\colon V \to V^*$, $\phi(w) = B(w, \cdot)$, is a linear isomorphism because B is nondegenerate. By the invariance of B and the dual action, ϕ is equivariant:

$$\phi(xw)(v) = B(xw, v) = -B(w, xv) = (x\phi(w))(v).$$

By construction $\phi(v_j') = v_j^*$, where (v_j^*) is the dual basis. Hence $\Omega_B = (\mathrm{id} \otimes \phi^{-1})\big(\sum_j v_j \otimes v_j^*\big)$. This is the image of the coevaluation tensor under an equivariant isomorphism. By Lecture 14, the coevaluation tensor is invariant and basis-independent. ∎

On $V = \mathbb{C}^N$ with $B(u, v) = u^{\mathsf{T}} v$, the Lie algebra $\mathfrak{so}_N$ consists of the matrices with $X^{\mathsf{T}} = -X$. The group $O_N = \{g : g^{\mathsf{T}} g = I\}$ acts on $V^{\otimes k}$ by $g^{\otimes k}$. The standard basis is orthonormal, so $e_i' = e_i$ and $\Omega = \sum_i e_i \otimes e_i$.

On $V = \mathbb{C}^{2n}$ with basis $e_1, \dots, e_n, f_1, \dots, f_n$, the symplectic form β is given by

$$\beta(e_i, f_j) = \delta_{ij} = -\beta(f_j, e_i), \qquad \beta(e_i, e_j) = \beta(f_i, f_j) = 0.$$

The partners of Lemma 27.6 are $e_i' = -f_i$ and $f_i' = e_i$. We use the sign-reversed invariant tensor

$$\Omega_{\mathrm{sp}} = -\Omega_\beta = \sum_{i=1}^{n} (e_i \otimes f_i - f_i \otimes e_i).$$

For even k and a perfect matching π of $\{1, \dots, k\}$ into pairs $a_j < b_j$, the *pair contraction* tensor Ω_π places the two factors of the jth copy of Ω in positions a_j, b_j. In other words, Ω_π comes from $\Omega^{\otimes k/2}$ by permuting tensor factors. Permuting factors is equivariant, so Ω_π is invariant. A *factor permutation* of a tensor in $V^{\otimes k} \otimes V^{*\otimes l}$ permutes the V-factors among themselves and the V^*-factors among themselves.

The *volume tensor* of $\mathbb{C}^N$ is

$$\mathrm{vol} = \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma)\, e_{\sigma(1)} \otimes \cdots \otimes e_{\sigma(N)}.$$

By the Leibniz formula and the analogous calculation for $\mathfrak{gl}_N$,

$$g^{\otimes N}\mathrm{vol} = \det(g)\,\mathrm{vol}, \qquad X\,\mathrm{vol} = (\operatorname{tr} X)\,\mathrm{vol} \quad \text{for } g \in GL_N,\ X \in \mathfrak{gl}_N.$$

So vol is $\mathfrak{sl}_N$-invariant. Antisymmetric matrices have trace zero, so vol is also $\mathfrak{so}_N$-invariant. A reflection $r \in O_N$ has $\det r = -1$ and sends vol to $-\mathrm{vol}$.

Theorem 27.7 (First fundamental theorems for tensors [C]). *Let V be the defining module and $k \geq 0$.*

(a) $(V^{\otimes k})^{O_N}$ *is zero for odd k. For even k it is spanned by the Ω_π.*

(b) For $N \geq 2$ (including the abelian case $N = 2$), $(V^{\otimes k})^{\mathfrak{so}_N}$ is the sum of $(V^{\otimes k})^{O_N}$ and the span of the factor permutations of $\mathrm{vol} \otimes \Omega^{\otimes (k-N)/2}$. *The second part occurs only if $k \geq N$ and $k - N$ is even.*

(c) $(V^{\otimes k})^{\mathfrak{sp}_{2n}}$ *is zero for odd k. For even k it is spanned by the symplectic Ω_π.*

(d) $(V^{\otimes k} \otimes V^{*\otimes l})^{\mathfrak{sl}_n} = 0$ *unless n divides $k - l$. First reorder tensor products into $V^{\otimes k} \otimes V^{*\otimes l}$. The invariants are spanned by factor permutations of products of pairings $\sum_i e_i \otimes e_i^*$ with $(k-l)/n$ copies of* vol *if $k \geq l$, or with $(l-k)/n$ copies of the dual volume tensor if $l \geq k$.*

For GL, O and Sp, see Goodman–Wallach, §§5.3.1–5.3.2. For the volume terms, see Fulton–Harris, Appendix F.1, Propositions F.10 and F.15. Polarisation gives the multilinear invariants. For SL_n, the dual volume identifies dual factors with exterior powers. We import these tensor forms. Passing to the Lie algebras of the connected groups SL_n, SO_N, Sp_{2n} preserves invariants: see Goodman–Wallach, §1.7.1, Proposition 1.7.7 and its proof.

The orthogonal volume term depends on the tensor degree. If $k < N$, there is no volume term, so $(V^{\otimes k})^{\mathfrak{so}_N}$ is spanned by pair contractions. We call $k < N$ the *stable range*. Linear independence of the Ω_π is a separate question. It fails when $\dim V$ is small. Exercise Sheet 27 shows this for the symplectic form on $\mathbb{C}^2$.

Since $V \cong V^*$ through B, Proposition 27.1 gives

$$\operatorname{End}_{\mathfrak{g}}(V^{\otimes d}) \cong \left(V^{*\otimes d} \otimes V^{\otimes d}\right)^{\mathfrak{g}} \cong (V^{\otimes 2d})^{\mathfrak{g}}.$$

The relevant tensor degree is $2d$, not d. In the orthogonal case, for $d = 2$ the three matchings of four points give the operators

$$\mathrm{id}, \qquad P(u \otimes v) = v \otimes u, \qquad E(u \otimes v) = B(u, v)\,\Omega.$$

In the symplectic case use $E(u \otimes v) = \beta(u, v)\Omega_{\mathrm{sp}}$.

Figure 27.1 shows the three orthogonal matchings. Positions 1, 2 are outputs; positions 3, 4 are inputs, identified with dual factors by B.

Lecture 22 used three maps of the same kind. Its intertwiners $\iota_{12}, \iota_{13}, \iota_{23} : V \to V^{\otimes 3}$ insert Ω_{sp} into two of the three output positions, and the remaining output position carries the input. These are the three matchings of four points, read with one input and three outputs.

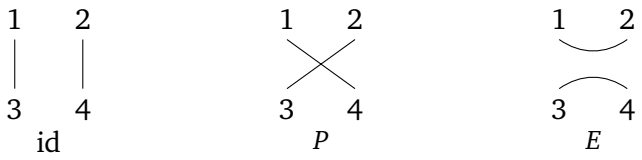


Figure 27.1: The three orthogonal matchings of four points, with their operators id, P and E. Each joining line pairs the indices at its endpoints.

Proposition 27.8 (Odd orthogonal commutants). *Let N be odd and $d \geq 1$. An operator on $V^{\otimes d}$ commutes with every $g^{\otimes d}$, $g \in SO_N$, if and only if it commutes with every $g^{\otimes d}$, $g \in O_N$.*

Proof. Since $SO_N \subseteq O_N$, one direction is clear. Conversely, let $g \in O_N$ with $\det g = -1$. Then

$$\det(-g) = (-1)^N \det g = 1,$$

so $h = -g \in SO_N$. Moreover, $g^{\otimes d} = (-1)^d h^{\otimes d}$, and a scalar multiple of $h^{\otimes d}$ commutes with everything that commutes with $h^{\otimes d}$. ∎

For even N and $2d \geq N$, the orthogonal invariant theorem gives further invariants. The tensor $\mathrm{vol} \otimes \Omega^{\otimes(2d-N)/2}$ is $\mathfrak{so}_N$-invariant, but a reflection negates it. Its operator lies in $\mathrm{End}_{\mathfrak{so}_N}(V^{\otimes d})$, not in the O_N-commutant. For $N = 4$ and $d = 2$ it is a nonzero multiple of the Hodge star on $\Lambda^2 V$, extended by zero on $\mathrm{Sym}^2 V$.

27.4 Worked example: the vector square

We now apply the commutant formula and the three matchings to one module.

Let $N \geq 5$, $\mathfrak{g} = \mathfrak{so}_N$ and $V = \mathbb{C}^N$. By Lecture 23,

$$V \otimes V = \mathbb{C}\Omega \oplus \Lambda^2 V \oplus \mathrm{Sym}^2_0 V,$$

where $\mathrm{Sym}^2_0 V$ is the kernel of the linear map induced by B on $\mathrm{Sym}^2 V$. The summands are irreducible for $N \geq 5$. Their dimensions

$$1, \qquad \frac{N(N-1)}{2}, \qquad \frac{N(N+1)}{2} - 1$$

are pairwise distinct, so the summands are pairwise nonisomorphic.

Degree two. $(V \otimes V)^{\mathfrak{g}} = \mathbb{C}\Omega$.

Degree three. Via B and Proposition 27.1, $(V^{\otimes 3})^{\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}(V, V \otimes V)$. The three dimensions above differ from N when $N \geq 5$, so Schur's lemma makes this space zero. This agrees with the stable range, since $3 < N$.

Degree four. Theorem 27.4 gives $\mathrm{End}_{\mathfrak{g}}(V \otimes V) \cong \mathbb{C}^3$. This matches the three matchings of four points, since $4 < N$ lies in the stable range. Write $B(\Omega)$ for the linear map induced by B applied to Ω. The operators $1 = \mathrm{id}$, P and E satisfy

$$B(\Omega) = \sum_i B(e_i, e_i) = N, \qquad E^2 = NE, \qquad PE = EP = E.$$

Here $E^2 = NE$ because $B(\Omega) = N$. Moreover, $PE = E$ because $P\Omega = \Omega$, and $EP = E$ because B is symmetric.

Figure 27.2 shows these relations as stacked matchings. We read each diagram from bottom to top, so the lower operator acts first. Stacking E over E joins the input arc of

the upper copy to the output arc of the lower copy. This closed loop is the number $B(\Omega)$. Stacking P over E puts a crossing on the output arc of E, which is P applied to Ω.

The symplectic form needs arrows, because β and Ω_{sp} change sign when their two factors are exchanged. Each arrow points from the first tensor factor to the second. The loop is then $\beta(\Omega_{\mathrm{sp}}) = 2n$. A crossing reverses the arrow, which gives the sign in $P\Omega_{\mathrm{sp}} = -\Omega_{\mathrm{sp}}$. Hence $E^2 = 2nE$ and $PE = -E$ in the symplectic case, and Exercise 27.6 proves these relations.

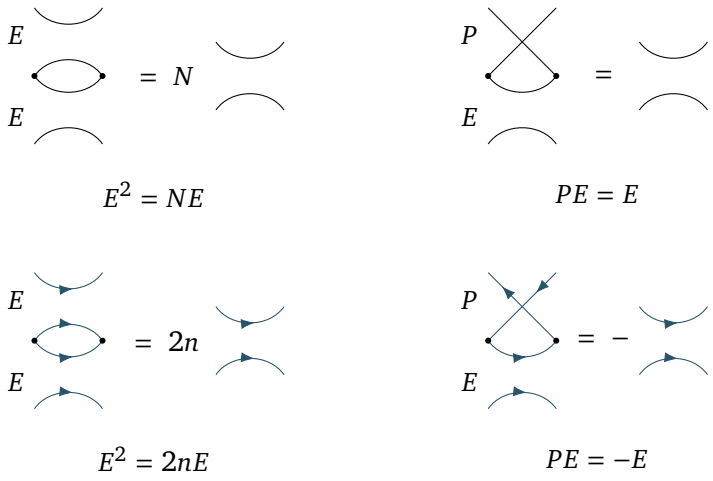


Figure 27.2: Composition of matchings, read upward, with every strand carrying V. In the upper row each arc $\cap$ is B and each arc $\cup$ is Ω. In the lower row each arc $\cap$ is β and each arc $\cup$ is $\Omega_{\mathrm{sp}} = -\Omega_\beta$, with arrows from the first tensor factor to the second.

The orthogonal idempotents are

$$p_0 = \frac{1}{N}E, \qquad p_- = \frac{1}{2}(1 - P), \qquad p_+ = \frac{1}{2}(1 + P) - \frac{1}{N}E.$$

Their images are $\mathbb{C}\Omega$, $\Lambda^2 V$ and $\mathrm{Sym}^2_0 V$. Under the isomorphism Ψ of Theorem 27.4, they correspond to the identity operators on the three one-dimensional multiplicity spaces.

The same method applies to the vector squares of $\mathfrak{sl}_n$ and $\mathfrak{sp}_{2n}$ from Lecture 18, Lecture 21 and Lecture 22. Exercise 27.6 constructs the symplectic projectors and compares the three cases.

A natural place to stop. We have identified intertwiners with invariant tensors and used them to split the vector square. The next section determines the representation ring from characters.

27.5 The representation ring

The first part of this lecture studied maps between modules. This section studies how modules multiply. A tensor product of modules decomposes into simple modules, and by Lecture 15 the character of a tensor product is the product of the characters. So characters turn tensor products into multiplication in $\mathbb{Z}[P]^W$, the ring of Weyl-invariant integer combinations of the exponentials e^μ, $\mu \in P$.

We record this multiplication in the representation ring. We then prove that this ring is a polynomial ring in the classes of the fundamental modules. Hence every class is a polynomial in finitely many classes. Exercise 27.5 carries this out for $\mathfrak{sl}_3$.

Write $\nu \le \mu$ if $\mu - \nu$ is a sum of simple roots with nonnegative integer coefficients. For $\lambda \in P_+$, the *orbit sum* is

$$o_\lambda = \sum_{\mu \in W\lambda} e^\mu.$$

By Lecture 9, each W-orbit in P meets P_+ exactly once. Therefore the o_λ, $\lambda \in P_+$, form a $\mathbb{Z}$-basis of $\mathbb{Z}[P]^W$.

Definition 27.9 (Representation ring)**.** The *representation ring* $K(\mathfrak{g})$ is the free abelian group with basis $[L(\lambda)]$, $\lambda \in P_+$. A module V has class

$$[V] = \sum_{\lambda \in P_+} \dim \mathrm{Hom}_{\mathfrak{g}}\big(L(\lambda), V\big)\, [L(\lambda)],$$

and multiplication extends $[L(\lambda)] \cdot [L(\mu)] = [L(\lambda) \otimes L(\mu)]$ bilinearly.

Complete reducibility and Lecture 14 give $[V] = [W]$ if and only if $V \cong W$, as well as $[V \oplus W] = [V] + [W]$. Since the tensor product distributes over direct sums, $[V][W] = [V \otimes W]$. Associativity, commutativity and the unit $[\mathbb{C}]$ follow from the module isomorphisms of Lecture 14.

By complete reducibility, every short exact sequence $0 \to U \to V \to W \to 0$ of modules splits. Hence $K(\mathfrak{g})$ agrees with the Grothendieck group defined by the relations $[V] = [U] + [W]$ for such sequences.

Lemma 27.10 (Unitriangular bases)**.** *Let $S \subseteq P_+$ be finite and downward closed in P_+. Let $\mathbb{Z}_S$ be the span of $\{o_\mu : \mu \in S\}$. Suppose that for each $\mu \in S$,*

$$a_\mu = o_\mu + \sum_{\nu \in S,\ \nu < \mu} c_{\mu\nu} o_\nu, \qquad c_{\mu\nu} \in \mathbb{Z}.$$

Then $(a_\mu)_{\mu \in S}$ is a $\mathbb{Z}$-basis of $\mathbb{Z}_S$.

Proof. List S so that $\nu < \mu$ implies that ν comes first. The matrix of the a_μ in the basis o_μ is then unitriangular with integer entries. Its determinant is 1, so its inverse has integer entries. ∎

By Lecture 9, every finite set $F \subseteq P_+$ lies in the finite downward-closed set

$$\{\nu \in P_+ : \nu \le \lambda \text{ for some } \lambda \in F\}.$$

Theorem 27.11 (Character isomorphism)**.** *The map $\mathrm{ch} \colon K(\mathfrak{g}) \to \mathbb{Z}[P]^W$, $[V] \mapsto \mathrm{ch}\, V$, is a ring isomorphism.*

Proof. By Lecture 15, characters are additive, multiplicative and Weyl invariant, and $\mathrm{ch}\, \mathbb{C} = 1$. So ch is a ring homomorphism into $\mathbb{Z}[P]^W$.

The weights of $L(\lambda)$ are $\le \lambda$, and λ has multiplicity one (Lecture 12). Weyl invariance therefore gives

$$\mathrm{ch}\, L(\lambda) = o_\lambda + \sum_{\nu \in P_+,\ \nu < \lambda} \dim L(\lambda)_\nu \, o_\nu.$$

Take any $x \in K(\mathfrak{g})$ and $f \in \mathbb{Z}[P]^W$. Choose a finite, downward-closed set S containing the support of x in the simple-class basis and that of f in the orbit-sum basis. The characters $\mathrm{ch}\, L(\mu)$, $\mu \in S$, lie in $\mathbb{Z}_S$, because S is downward closed.

Lemma 27.10 shows that ch maps the span of $\{[L(\mu)] : \mu \in S\}$ bijectively onto $\mathbb{Z}_S$. Hence $\mathrm{ch}\, x = 0$ forces $x = 0$, and $f = \mathrm{ch}\, y$ for some y in the span of the $[L(\mu)]$, $\mu \in S$. ∎

Let $r = \operatorname{rank}\mathfrak{g}$ and $\chi_i = \operatorname{ch} L(\omega_i)$. For $\mu = \sum_i \mu_i \omega_i \in P_+$, put

$$\chi^\mu = \prod_{i=1}^{r} \chi_i^{\mu_i}.$$

Theorem 27.12 (Fundamental characters)**.** *The ring $\mathbb{Z}[P]^W$ is generated by $\chi_1, \dots, \chi_r$, which are algebraically independent. Hence $K(\mathfrak{g}) \cong \mathbb{Z}[x_1, \dots, x_r]$, with $x_i \mapsto [L(\omega_i)]$.*

Proof. The element χ^μ is the character of $T_\mu = \bigotimes_i L(\omega_i)^{\otimes \mu_i}$. A weight of T_μ is a sum $\nu = \sum_f \nu_f$ over the factors, with each ν_f at most the highest weight of its factor. Thus

$$\mu - \nu = \sum_f (\text{highest weight of } f - \nu_f) \ \geq 0.$$

Equality holds only if every ν_f is highest. Highest weight spaces are one-dimensional, so μ has multiplicity one. Since χ^μ is also Weyl invariant,

$$\chi^\mu = o_\mu + \sum_{\nu \in P_+,\ \nu < \mu} c_{\mu\nu} o_\nu.$$

For a finite, downward-closed S, Lemma 27.10 makes $(\chi^\mu)_{\mu \in S}$ a basis of $\mathbb{Z}_S$. Every element of $\mathbb{Z}[P]^W$ lies in some $\mathbb{Z}_S$, so the monomials χ^μ span. This proves generation.

A polynomial relation among the χ_i is a linear relation among monomials $\prod_i \chi_i^{a_i} = \chi^{\sum a_i \omega_i}$. Distinct exponent vectors give distinct μ, because the ω_i form a basis of P. Finitely many of these monomials lie in one basis $(\chi^\mu)_{\mu \in S}$, so they are linearly independent. Theorem 27.11 transfers the result to $K(\mathfrak{g})$. ∎

For example, for $\mathfrak{sl}_2$ let $x = [V_1]$. Clebsch–Gordan gives

$$x[V_m] = [V_{m+1}] + [V_{m-1}] \quad \text{for } m \geq 1.$$

The cases $m = 1$ and $m = 2$ give

$$[V_2] = x^2 - 1, \qquad [V_3] = x[V_2] - [V_1] = x^3 - 2x.$$

As a preview, Lecture 28 uses a second algebra of Weyl invariants. It is the algebra $\operatorname{Sym}(\mathfrak{h})^W$ of W-invariant polynomial functions on $\mathfrak{h}^*$, with complex coefficients. The ring $\mathbb{Z}[P]^W$ of this section has integer coefficients, and its elements are combinations of exponentials. These are different algebras. Lecture 28 does not use $K(\mathfrak{g})$, and no result there depends on this section.

Self-checks

(1) What is $\dim \mathrm{End}_{\mathfrak{sl}_2}(V_1^{\otimes 3})$?
(2) For $N \geq 2$, when is $(V^{\otimes 3})^{\mathfrak{so}_N} \neq 0$?
(3) Express $[V_4]$ in $K(\mathfrak{sl}_2) = \mathbb{Z}[x]$.

Answers. (1) $V_1^{\otimes 3} \cong V_3 \oplus 2V_1$, so the dimension is $1^2 + 2^2 = 5$. (2) Only for $N = 3$. Odd degree has no O_N part, and the volume part needs $3 \geq N$ with $3 - N$ even. (3) $[V_4] = x[V_3] - [V_2] = x^4 - 3x^2 + 1$. The dimensions check: $16 - 12 + 1 = 5$.

Summary

Intertwiners are invariant tensors. Multiplicity spaces determine commutants. Classical invariant tensors are built from pairings and volume tensors.

Characters identify $K(\mathfrak{g})$ with $\mathbb{Z}[P]^W$, a polynomial ring in the fundamental characters. Lecture 28 turns to the centre of $U(\mathfrak{g})$.

Reading

Fulton–Harris, Appendix F.1, and Goodman–Wallach, §5.3, give the invariant theorems cited above. Kirillov, §8.6, treats multiplicities. Exercise Sheet 27 gives further calculations.